ARTUR KASHPERSKIY

Software Engineer

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sm5art

EXPERIENCE

Software Engineer
Autodesk

Sep 2022 – Current | San Francisco, CA

- Joined User Profile team in Identity/Licensing working on infrastructure for user profile platform across Autodesk products (authz/authn). profile.autodesk.com.
- Built up Observability dashboards/alarms on AWS Cloudwatch and Splunk tracking latencies, errors, and call counts on our various API services/operations.
- Worked on various infrastructure/code and internal library optimizations that improved our API latency from 1.5s (avg) when joining to now averaging about 0.3s in response time while also adding more features to the APIs. Rolled out rate-limiting features for users of our API which improved stability and performance. Improved library for clients for Redis cache layer and https requests through various environment tests.
- Included on-call support rotation for a two week period every 8 weeks for our team supporting our APIs using various internal tools for Dynatrace alerting to PagerDuty and other tools like Splunk and Cloudwatch for diagnosis/monitoring; worked on setting up wikis for run-books for alarming/diagnosis.
- GraphQL, nodeJS, serverless, AWS SQS/S3/Lambda/DynamoDB, AWS Kinesis/Cloudwatch, Redis, Terraform, Splunk, PagerDuty, Dynatrace

Software Engineer Intern
Autodesk

June 2021 – September 2021 | San Rafeal, CA

- Worked with User Profile team to build front end for new landing page for when users first hit profile.autodesk.com (to manage a users profile).
- Worked with designers to build a few reusable React components and microfrontend that supports fast loading of the new landing page. Learned JS technologies for localization, GraphQL queries and microfrontend routing.
- GraphQL, React, jenkins, nodeJS, Dynatrace, Adobe AEM

Software Engineer Intern
Valravn Capital

January 2021 – March 2021 | NY, Remote

- Built up new infrastructure and helped team with moving and consolidating some fundamental macroeconomic data fetching processes for research purpose to new Airflow infrastructure to help reduce costs spent on AWS.
- Built an experimental fintech full stack web application for optimizing and simulating many different portfolios with different exposure options using plotly Dash.
- Apache Airflow, Python, AWS

Software Engineer Intern
Heali.ai

July 2019 – October 2019 | Santa Monica, CA

- Built APIs with Express.js and serverless to support feature for mobile app showing diet and nutrition information when scanning food product barcodes with camera.

SKILLS

Python | JavaScript | Java | C | C++ | C# | MATLAB | React | GraphQL | React Nativ | Docker | Kubernetes | Unreal Engine | Unity | MongoDB | AWS EC2 | API Gateway | Lambda | S3 | MariaDB | Redis | Memcached | Terraform

PROJECTS

demix

- A web tool which allowed music producers to upload an audio file and receive vocals and instrumental individually separated from the file. This project saved music producers time and effort by giving them a ml-based approach to remix songs instead of manual equalisation. Launched to production using AWS EC2 and docker for deployment for 2 months gaining 8000 WAU (had to shutdown due to high cost). Stack: python Flask server, GatsbyJS frontend, tensorflow.
- https://github.com/sm5art/demix
- https://github.com/sm5art/demix-frontend

genetic pong

- Forked a friend's pong game written in the python package pygame from his GitHub and wrote a genetic algorithm from scratch on top in python that used an unsupervised fitness heuristic to optimise for the fittest pong "brain". The training sequence took roughly 30 minutes or 30 generations to converge and resulted in an unbeatable AI player.
- YouTube Training Demo
- https://github.com/sm5art/genetic-pong

EDUCATION

B.S. in Applied Physics
University of Washington

Sept 2017 – December 2022

Notable coursework: Algorithms and Data Structures, Vector Calculus/Complex Analysis, Probability I, Artificial Intelligence, MATLAB for Numerical Analysis, Quantum Mechanics, Digital/Analog Electronics

- Built front end functionality for food product barcode scanning with Swift native OCR plugins in React Native App.
- Added some new infrastructure for bug reporting (Phabricator) and integrated it with a shake to bug report frontend feature to the app.
- React Native, serverless, nodeJS, Apache Phabricator, AWS, Swift

Software Engineer Intern

Nipun Capital

June 2017 – September 2017 Foster City, CA

- Built up data pipeline for alpha generation and web scraping sources in emerging markets (exchange sites) using python and Apache Airflow (one job required to break old text capchas using CNNs). Moved old alpha generation cronjobs to new Airflow infrastructure.
- Worked on a small NLP research project investigating sentiment analysis of conference call transcripts (financial sentiment dictionary and word2vec methods) and backtested using internal tools.
- python, Google Cloud, Apache Airflow, tensorflow

Software Engineer Intern

Minted

June 2016 – August 2016 San Francisco, CA

- Wrote unit/integration tests improving coverage of marketing insert logic in Minted fulfillment state machine and fixed bugs.
- Built an API endpoint and front end for an analytics dashboard used as a productivity tracking tool for our stationary design auditing team.
- BackboneJS/ReactJS, Python, MySQL, Vagrant, puppeteer